Comparison of interpolation methods in time and frequency domain for the estimation of harmonics and interharmonics according to IEC standard

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***Abstract*— The fast Fourier transform (FFT) is the most commonly used technique in power systems signals analysis. However, it is known that if the signal is not synchronized or in the presence of interharmonics, the spectral leakage may occur. To overcome this issue, several techniques are found in literature. The present paper will show a comparison between results of calculating the IEC 61000-4-7 harmonics and interharmonics groups and subgroups using two of these techniques, one that uses frequency domain interpolation and other that uses time domain interpolation.**

***Index Terms*— Fast Fourier Transform, frequency domain interpolation, frequency variation, time domain interpolation, synchronization.**

I. Introduction

The proliferation of electronic devices either as load or generator unity has increased the harmonic distortion in power system as well the injection of interharmonic components. As defined in IEC 61000-2-1 [1], interharmonics are those components with frequencies lying between harmonics of the voltage or current waveforms and are not an integer multiple of the system fundamental frequency. Interharmonics are generated by the intermodulation of the fundamental and harmonic components with other frequency component due to the intrinsic device working cycle. It can be observed in a large number of loads, such as static frequency converters, cycloconverters, induction motors, arc furnaces and loads not pulsated synchronously pulsated with the fundamental frequency [2]. In addition, the fundamental frequency of the system may vary because of the power mismatch between the generation and the load demand.

The most commonly technique used for harmonic and interharmonic analysis in power systems is the fast Fourier transform (FFT). The direct implementation of this technique may lead to erroneous results due to spectral leakage and picket fence effect in two situations: in the presence of interharmonics and when the fundamental frequency either varies or is different of the supposed one [3].

To address the interharmonic issue, the IEC 61000-4-7 [4] provides a general guide for harmonic and interharmonic measurements up to 9 kHz by the grouping method. A 200 ms rectangular window is used that corresponds to 12 cycles of 60 Hz or 10 cycles of 50 Hz, giving a spectral resolution of 5 Hz.

For the leakage problem, the IEC 61000-4-7 suggests the application of Hanning window instead of the rectangular one. However, when there is a frequency deviation, that is not sufficient and other techniques, like interpolation, should be applied. The interpolation can be done either in time domain, synchronizing the signal before the FFT calculation, or in frequency domain, implemented after the FFT performed over the unsynchronized signal.

The time domain interpolation approximates the signal with a continuous function and resamples the approximated function at appropriate time moments. It can be done either offline (in a block manner) [5]-[8] or online (sample by sample) [9]-[10]. In this way, these methods synchronize the signal before the FFT calculation. Frequency domain methods operates on the FFT results, correcting the values of the amplitude, phase and frequency of each harmonic component [2], [11]. Furthermore, frequency domain methods assume a constant frequency inside the analyzed window while time domain methods utilize the instantaneous frequency, avoiding spectral leakage in frequency variation situations.

This paper will show a comparison of the results of two methods, one for time domain interpolation and other one for frequency domain interpolation in three scenarios: frequency deviation, ramp frequency variation and sinusoidal frequency variation, highlighting the advantage of each method. The

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time and frequency domain methods are presented in [10] and [2], respectively.

The paper is divided as follows: Section II describes the methods that were used for time and frequency domain interpolation and the IEC 61000-4-7 group methodology; Section III presents the results of the application of these methods as well as the discussions about it. Finally, Section IV presents the conclusions about this work.

II. Utilized Methods

A. Time Domain Interpolation

The time domain interpolation corresponds to the process of making the signal synchronous. It can be done either online or offline by the application of several methods. Among them, the B-spline interpolation as shown in [10] is used. This process is defined as shown in the diagram of the Fig.1.

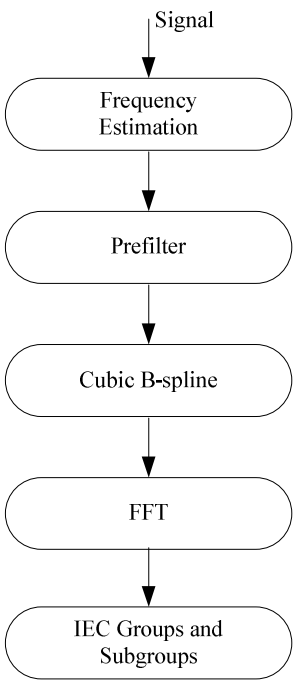

Fig.1. Diagram of the method using time interpolation.

Analyzing Fig.1, one can notice a five-step process. The first step, the frequency estimation of the signal is performed and its value is used in the time domain interpolation. The second and the third steps are the time domain interpolation. First we have a Pre-filter and then a Cubic B-spline. After these blocks, the signal becomes synchronous and the FFT can be applied. In the fourth block, the FFT of the synchronized signal is evaluated and its results are used in the sixth block for the calculation of the IEC harmonics and interharmonics groups and subgroups.

In the next section, the results of the time domain interpolation method will be shown.

B. Frequency Domain Interpolation

The frequency domain interpolation method implemented in this paper is described in [2]. The global diagram of the methodology is shown in Fig.2. First, the input signal is sampled and windowed by a predefined window and then its FFT is calculated. Then, the interpolation is performed in order to evaluate the true value for the bins matching the harmonic components. Once these values are obtained, the effect of the harmonics can be removed from the spectrum obtained by the FFT and then, this spectrum will have only the interharmonic content. Based on the corrected harmonic values and the interharmonic spectrum, the IEC Harmonic and Interharmonic groups and subgroups are calculated.

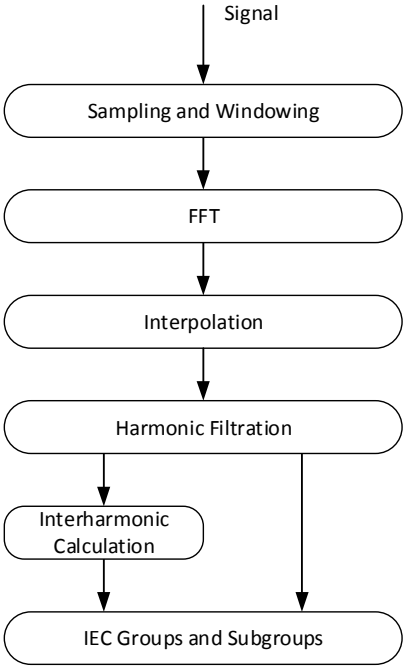

Fig.2. Diagram of the method using frequency interpolation.

It is possible to notice that the time domain interpolation provides better results than the frequency domain interpolation in the presence of frequency variation.

C. IEC 61000-4-7

The IEC 61000-4-7 describes the application of groups and subgroups of harmonics and interharmonics obtained from frequency spectrum. This methodology has the aim of grouping the energy of the components scattered along the spectrum in order to obtain the value of each harmonic and interharmonic component.

This approach is simple and it is presented, for a 60 Hz system, in Fig.1, for harmonics, and in Fig.2, for interharmonics.

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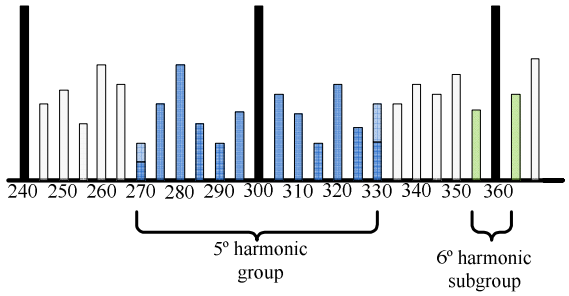

Fig.3. Harmonics groups and subgroups.

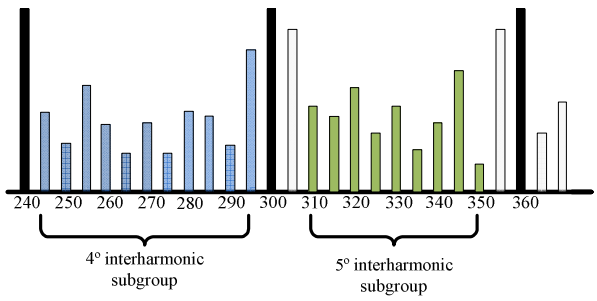

Fig.4. Interharmonic group and subgroup.

In Fig.3, the application for a harmonic group and subgroup is presented. Thus, it can be noted that for the harmonic group calculation, the harmonic bin and the twelve adjacent bins (six of each side) are considered. It should be noted that for the sixth bin in each side, only the half of its magnitude is considered because it is taking into account on the calculation of two adjacent harmonic group. For the subgroup, only the bin of the harmonic and the two adjacent ones are considered. Fig.3 shows the calculation for both the fifth harmonic group and the sixth harmonic subgroup.

Fig.4 shows the application of the interharmonic groups and subgroups. For the calculation of the group, all the bins between two consecutive harmonics (in this case the fourth and the fifth) are considered. For the subgroups, all the bins, except the bins adjacent to the harmonics, are considered.

The equations that describe the harmonic and interharmonic group and subgroup calculations are:

$$G_{g,h}^2 = \left\{ \frac{C_{k_h-6}^2}{2} + \sum_{i=-5}^{5} C_{k_h+i}^2 + \frac{C_{k_h+6}^2}{2}, \right. \quad (1)$$

$$G_{sg,h}^2 = \sum_{i=-1}^{1} C_{k_h+i}^2 \quad (2)$$

$$G_{ig,h}^2 = \sum_{i=1}^{11} C_{k_h+i}^2, \quad (3)$$

$$G_{sig,h}^2 = \sum_{i=2}^{10} C_{k_h+i}^2, \quad (4)$$

where the parameters $G_{g,h}^2$, $G_{sg,h}^2$, $G_{ig,h}^2$, $G_{sig,h}^2$, are the quadratic values of the harmonic group, harmonic subgroup, interharmonic group and interharmonic subgroup, respectively. $C_{k_h+i}^2$ represents the quadratic value of each bin of index k_h+i, where k_h is the index of the hth harmonic. For the application of the Hanning window, a factor of $\sqrt{2/3}$ should be used, in order to obtain the correct value of the magnitude.

It was be seen on this section that when the frequency is fixed, but different from the supposed one, the frequency domain interpolation performs well, but when the frequency varies, the results are not satisfactory.

III. Results and Discussions

For the tests shown in this section, a signal that contains all odd harmonics up to 50th order and two interharmonics components with frequencies equal to 83.23 Hz and 442.8 Hz of magnitude 0.1, are used in order to highlight the effects of the interpolation methods, a fundamental frequency deviation was applied, considering a 60 Hz system. The magnitude values are 1 pu for the fundamental component and 0.3 pu for the harmonics up to order 40. For the other harmonics, the value of 0.1 is used. An ideal frequency estimator was considered in the time domain interpolation process. Rectangular window and Hanning window were used in order to compare their results. Only the IEC 61.000-4-7 subgroups were calculated, since the presence of an interharmonic affects the group calculation hardly.

The first test was performed considering a constant frequency deviation. The fundamental frequency was turned to 59 Hz. The results for both the harmonics and interharmonics subgroups are shown in Fig.5 and Fig.6, respectively.

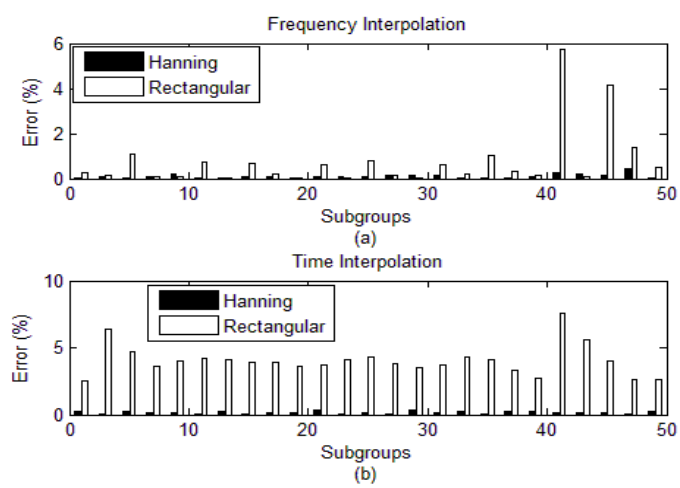

Fig.5. Harmonic Subgroups: (a) time domain interpolation (b) frequency domain interpolation.

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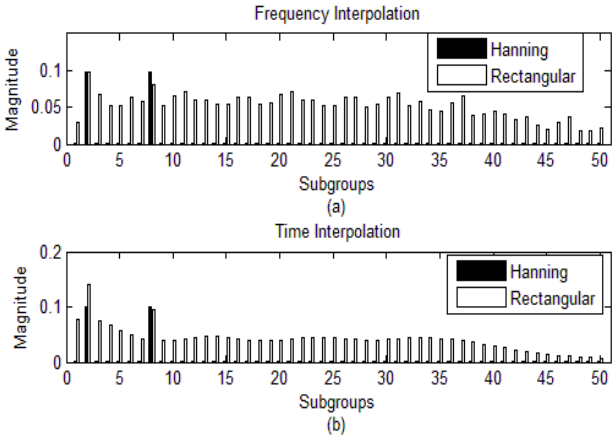

Fig.6. Interharmonic Subgroups: (a) frequency domain interpolation (b) time domain interpolation.

On Fig.5(a) and Fig.6(a) the results of the application of the frequency domain interpolation are shown for the harmonic subgroups and for the interharmonic subgroups respectively. Similarly, the

Fig.5(b) and Fig.6(b) show the results for the interharmonic subgroups in the case where the time domain interpolation is used. In this scenario, it can be noted that the Hanning window performs better than the Rectangular window for both interpolation methods, and the errors obtained with time domain interpolation and with frequency domain interpolation are very similar for both harmonic and interharmonic subgroups. The results are very similar for the two methods, because the fundamental frequency, despite being different of 60 Hz, was constant inside the FFT window. Hence, when the correction performed by the frequency domain interpolation is applied, the synchronization is longer necessary.

The next step presents the results of the two interpolation methods in the scenario of frequency variation. For it, a fundamental frequency of 60 Hz was considered and two kinds of variation were adopted, ramp variation and sinusoidal variation. The frequency variation partner for these two cases are shown in Fig.7. In the ramp variation, the frequency starts in 60 Hz and in two seconds reaches 61 Hz. In the sinusoidal variation, the frequency varies with 8 seconds period and 1 Hz of amplitude.

Fig.8 shows the harmonic subgroups for the frequency variation, in ramp considering the application of time and frequency domain interpolation. Fig. 9 shows the same for the interharmonic subgroups. It can be noted that, once again, the Hanning window performed better than the Rectangular one and, in this case, better results were obtained with time domain interpolation. Its worth monitoring that the leakages in interharmonic subgroups were lower for time domain interpolation. This is due to the synchronization performed in the signal at the time domain.

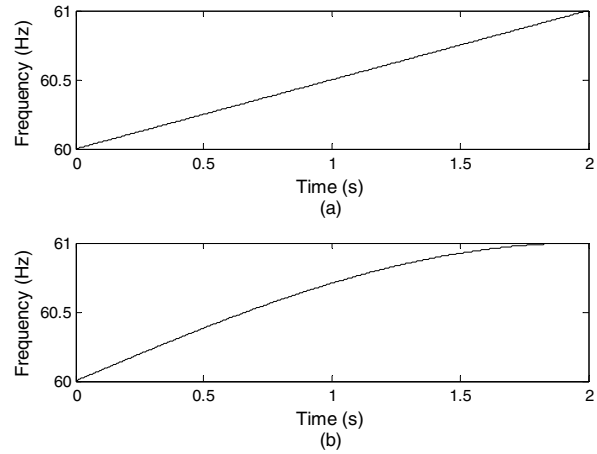

Fig.7. Frequency: (a) ramp variation (b) sinusoidal variation.

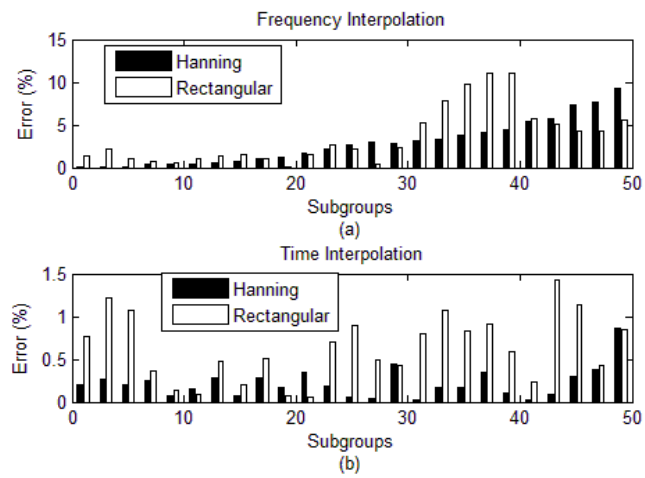

Fig.8. Harmonics subgroups, (a) frequency interpolation (b) time interpolation.

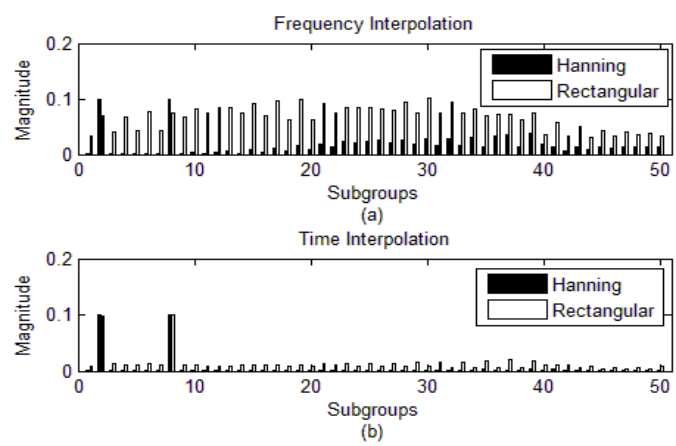

Fig. 9. Interharmonics subgroups, (a) frequency domain interpolation (b) time domain interpolation.

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For the sinusoidal variation, the results are shown in Fig.10 for the harmonic subgroups and in Fig.11 for the interharmonic subgroup.

Again, it can be noted that the time domain interpolation over performed the frequency domain interpolation. This is due the fact that the time domain interpolation synchronizes the signal before the FFT calculation and the frequency domain interpolation does not. In frequency variation scenario, the frequency of the signal varies in the FFT window, so the results of the FFT cannot be corrected as in the case of frequency domain interpolation, while the time domain interpolation uses the instantaneous frequency of the signal.

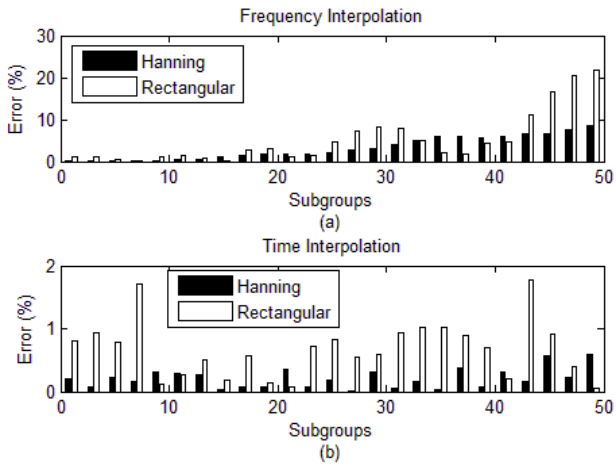

Fig.10. Harmonics subgroups, (a) frequency domain interpolation (b) time domain interpolation.

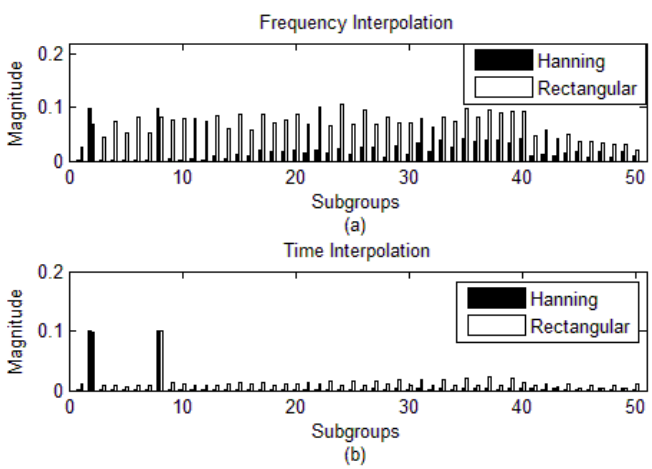

Fig.11. Interharmonics subgroups, (a) frequency domain interpolation (b) time domain interpolation.

IV. Conclusion

This paper presented two methods for compute the IEC 61000-4-7 harmonics and interharmonics subgroups (or group) based on interpolation techniques, both in frequency and time domain. The advantage of these techniques is that some degree of frequency deviation is accepted as the methods set themselves to work appropriately under incoherent sampling. The spline based time interpolation method, together with Hanning window, led to better results in almost situations. However, time interpolation needs the frequency information, which can be obtained by a PLL device not implemented in this paper. Frequency interpolation estimates the frequency, as an intrinsecal. However, if the frequency is timing varying, the frequency domain interpolation does not work appropriately. Future work will address the PLL frequency estimation, as a pre-processing for the spline interpolator.

Acknowledments

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